

Lyapunov approach to study stability of a particular class of triangular hyperbolic systems

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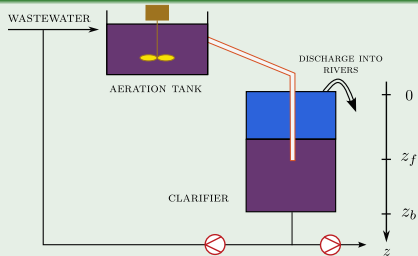
MATHIEU BAJODEK

In Collaboration with:

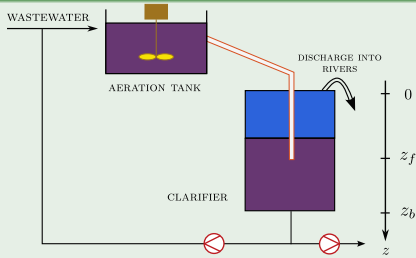
JEAN AURIOL (L2S)



Framework



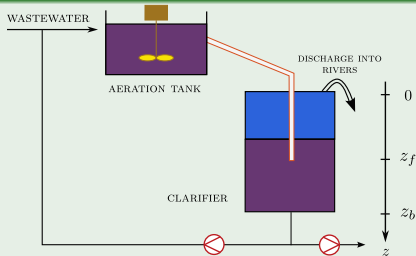
Framework



1D-Dynamic Model of the Clarifier (Two-Phase flow)

State variables: ε_s (particles density), $f_s := \varepsilon_s v_s$ (momentum) with v_s the particles velocity

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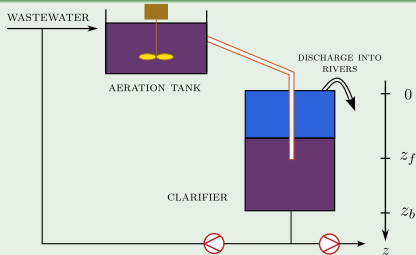


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$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \frac{f_{1s}}{\rho_s} \delta_{z_f} \\ \partial_t f_s + \partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right) = \underbrace{\varepsilon_s g \left(1 - \frac{\rho_l}{\rho_s} \right)}_{\text{Apparent weight}} + \underbrace{\frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)}}_{\text{Friction}} + \underbrace{\frac{f_{2s}}{\rho_s} \delta_{z_f}}_{\text{Feed}} \end{cases}$$

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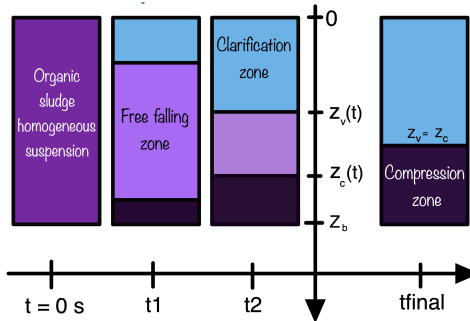
Goal 2: Prove the stability of the controlled system

OUTLINE OF TALK

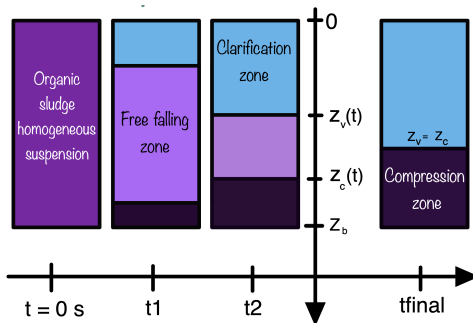
- 1 BATCH COLUMN FRAMEWORK
- 2 THROUGH A SIMPLIFIED CLARIFIER
- 3 NUMERICAL RESULTS
- 4 CONCLUSIONS AND PERSPECTIVES

BATCH COLUMN FRAMEWORK

SETTLING IN A BATCH COLUMN



SETTLING IN A BATCH COLUMN



N. CHASSIN, C. VALENTIN, J.-M. CHOUBERT, F. COUENNE, C. JALLUT, 2022, *How to improve modeling of urban sludge settling in a batch column*, Note interne



C. VALENTIN, F. LAGOUTIÈRE, J.-M. CHOUBERT, F. COUENNE, C. JALLUT, 2023, *Knowledge-based model and simulations to support decision making in Wastewater Treatment*, Processes PROCEEDINGS 33rd European Symposium on Computer Aided Process Engineering (ESCAPE33), (6p), Athens. Greece

SYSTEM ANALYSIS

Governing equations

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$D =]0, z_b[\times]0, T[$ / Find $\varepsilon_s, f_s : D \longrightarrow \mathbb{R}$ such that :

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ρ_l : water density ($:= 1000 \frac{\text{kg}}{\text{m}^3}$)

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$\sigma_0, n_s, n_r, \varepsilon_c, \tau, \rho_s$ (parameters)

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⇒ Strictly hyperbolic system

$$\text{Sp}(\mathbf{A}) = \{ v_s, v_s \} ; \text{ si } \varepsilon_s \leq \varepsilon_c$$

⇒ Weakly hyperbolic system

SYSTEM ANALYSIS

Map to follow

- Search of steady-states: $\mathbf{Y}(t, z) = \mathbf{Y}^*(z), \forall t \geq 0$
 - Linearize the system around it
- Find stability conditions for the linearized system
- Concluding stability of the steady-states for the nonlinear system may require additional conditions because of the infinite dimensional framework

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Source term implies **non-uniform steady-states**

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Notations:

$$K = g(1 - \frac{\rho_l}{\rho_s}) \quad / \quad \gamma = \frac{\tau}{\rho_s} \quad / \quad \beta = \frac{n_r - 1}{3 - n_r}$$

SYSTEM ANALYSIS

Find the steady-states: solve the following system of ODEs:

$$\begin{cases} \frac{df_s^*}{dz} = 0, \\ \frac{d\left(\frac{f_s^{*2}}{\varepsilon_s^*} + \frac{\sigma_\varepsilon(\varepsilon_s^*)}{\rho_s}\right)}{dz} = K\varepsilon_s^* - \gamma \frac{(\varepsilon_s^*)^\beta f_s^*}{1 - \varepsilon_s^*}, \\ f_s^*(0) = 0, \quad f_s^*(z_b) = 0 \end{cases}$$

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Set of steady-states

$$f_s^*(z) = 0, \quad \forall z \in [0, z_b]$$

$$\varepsilon_s^*(z) = \begin{cases} 0, & 0 \leq z \leq z_c \\ \left(\varepsilon_c^{n_s-1} + \frac{(n_s-1)K}{M}(z - z_c)\right)^{\frac{1}{n_s-1}}, & z_c < z \leq z_b \end{cases}$$

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How to determine z_c ? (to fully define the set of steady-states)

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Conservation principle of mass

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which leads to solving **for** z_c the following non-linear algebraic equation:

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RESULT

For any given initial condition $\mathbf{U}^0$ and fixed set of parameters $(\sigma_0, n_s, n_r, \varepsilon_c, \tau, \rho_s)$,

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RESULT

For any given initial condition $\mathbf{U}^0$ and fixed set of parameters $(\sigma_0, n_s, n_r, \varepsilon_c, \tau, \rho_s)$, we can determine analytical expressions of steady-states

NUMERICAL SIMULATION (GODUNOV-TYPE FINITE VOLUME SCHEME)

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Also desired steady-state after design of boundary controller for the **continuous framework settling**

LINEARIZATION AROUND STEADY-STATES

System in *primitive* variables:

$$\begin{cases} \partial_t \varepsilon_s + \partial_z(\varepsilon_s v_s) = 0, \\ \partial_t v_s + \partial_z\left(\frac{v_s^2}{2}\right) + \frac{1}{\rho_s \varepsilon_s} \partial_z \sigma_e(\varepsilon_s) = K - \gamma \frac{\varepsilon_s^\beta v_s}{1 - \varepsilon_s} \end{cases} \quad (1)$$

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Let us define the deviations of the states with respect to steady-states:

$$E_s := \varepsilon_s(t, x) - \varepsilon_s^*(x), \quad V_s := v_s(t, x) - v_s^*(x)$$

LINEARIZATION AROUND STEADY-STATES

System in *primitive* variables:

$$\begin{cases} \partial_t \varepsilon_s + \partial_z(\varepsilon_s v_s) = 0, \\ \partial_t v_s + \partial_z\left(\frac{v_s^2}{2}\right) + \frac{1}{\rho_s \varepsilon_s} \partial_z \sigma_e(\varepsilon_s) = K - \gamma \frac{\varepsilon_s^\beta v_s}{1 - \varepsilon_s} \end{cases} \quad (1)$$

For smooth solutions, system (1) is equivalent to the system in *conservative* variables, *but not for weak solutions*

Let us define the deviations of the states with respect to steady-states:

$$E_s := \varepsilon_s(t, x) - \varepsilon_s^*(x), \quad V_s := v_s(t, x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \mathbf{Y} + \mathbf{A}^*(z) \cdot \partial_z \mathbf{Y} + \mathbf{B}^*(z) \cdot \mathbf{Y} = 0$$

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LINEARIZATION AROUND STEADY-STATES

Result

When the system is strictly hyperbolic ($z_c < z \leq z_b$), Riemann Invariants are:

$$R_1^* = V_s + \frac{\sqrt{\frac{\sigma'_e(\varepsilon_s^*)}{\rho_s}}}{\varepsilon_s^*} E_s, \quad R_2^* = V_s - \frac{\sqrt{\frac{\sigma'_e(\varepsilon_s^*)}{\rho_s}}}{\varepsilon_s^*} E_s$$

In this case, besides being linear, the system can also be put into *characteristic form*

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And then how to deal simultaneously with both cases (handling the strictly-weakly switch) ?

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How to do when the system is weakly hyperbolic ? No diagonalizing transformation available

And then how to deal simultaneously with both cases (handling the strictly-weakly switch) ? Looking for a C^1 transformation that rather leads to a **diagonal triangular form**

THROUGH A SIMPLIFIED CLARIFIER

STABILITY ANALYSIS OF A SIMPLIFIED CLARIFIER: A TRIANGULAR HYPERBOLIC SYSTEM

$$\begin{cases} \partial_t \phi(t, x) = \Lambda(x) (A(x) \partial_x \phi(t, x) + B(x) \phi(t, x)), & x \in (0, 1), t \geq 0 \\ \phi(t, 1) = D \phi(t, 0), \\ \phi(0, x) = \phi^0(x) \end{cases} \quad (2)$$

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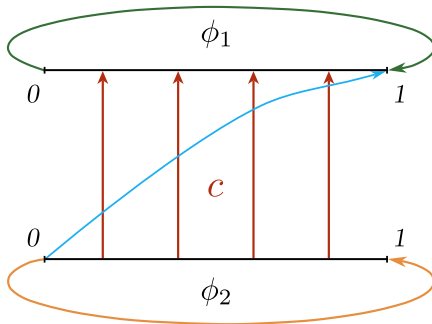
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Schematic representation of the system

STABILITY ANALYSIS OF A SIMPLIFIED CLARIFIER: A TRIANGULAR HYPERBOLIC SYSTEM

State space

$X = L^2((0, 1); \mathbb{R}) \times H^1((0, 1); \mathbb{R})$ equipped with the norm:

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Exponential stability

The origin of system (2) is δ -exponentially stable if there exists an overshoot $\kappa \geq 1$ and a decay rate $\delta > 0$ such that for any initial condition $\phi^0 \in X$:

$$\|\phi(t; \phi^0)\|_X^2 \leq \kappa e^{-\delta t} \|\phi^0\|_X^2, \quad \forall t \geq 0 \quad (3)$$

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Sufficient conditions that ensures exponential stability of this system ?

To answer, stability analysis is performed. The study will be divided into two cases: Without and with source term B

FIRST CASE: $B = 0$

Assumption 1

The function λ_2 and the matrix D satisfy the following inequality:

$$\max(D_{11}^2, D_{22}^2) < 1 - \frac{\overline{\lambda_2'}}{\underline{\lambda_2}}.$$

Theorem 1

Let Assumption 1 holds and let $\nu := \frac{\overline{\lambda_2'}}{\underline{\lambda_2} - \overline{\lambda_2'}}$, $\bar{\nu} := \frac{1 - D_{22}^2}{D_{22}^2}$. Then, for any $\delta < \min\{\frac{\bar{\nu}\lambda_1}{1+\bar{\nu}}, \frac{\bar{\nu}\lambda_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\lambda_2}{(1+\bar{\nu})(1+\nu)}\}$, the origin of system (2) is δ -exponentially stable.

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Proof: The method used to prove Theorem 1 relies on the construction of a Lyapunov functional

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Consider the following candidate functional:

$$\mathcal{V}(\phi) = \mathcal{V}_1(\phi) + \alpha_{20}\mathcal{V}_{20}(\phi) + \alpha_{21}\mathcal{V}_{21}(\phi)$$

With

$$\mathcal{V}_1(\phi) = \int_0^1 \frac{1+\mu x}{\lambda_1(x)} \phi_1^2(x) dx,$$

$$\mathcal{V}_{20}(\phi) = \int_0^1 \frac{1+\mu x}{\lambda_2(x)} \phi_2^2(x) dx,$$

$$\mathcal{V}_{21}(\phi) = \int_0^1 \frac{1+\mu x}{\lambda_2(x)} (\partial_x \phi_2)^2(x) dx$$

and scalars $\mu, \alpha_{20}, \alpha_{21} > 0$.

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Recall that $\mathcal{V}$ induces a norm on X that is equivalent to the state norm $\|\cdot\|_X$

FIRST CASE: $B = 0$

Time derivative of each term of $\mathcal{V}$ along C^1 solutions of system (2)

$$\dot{\mathcal{V}}_{20}(\phi) = 2 \int_0^1 (1 + \mu x) \phi_2(x) \partial_x \phi_2(x) dx,$$

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Time derivative of each term of $\mathcal{V}$ along C^1 solutions of system (2)

$$\begin{aligned}\dot{\mathcal{V}}_{20}(\phi) &= 2 \int_0^1 (1 + \mu x) \phi_2(x) \partial_x \phi_2(x) dx, \\ &= -\mu \int_0^1 \phi_2^2(x) dx + (1 + \mu) \phi_2^2(1) - \phi_2^2(0)\end{aligned}\tag{4}$$

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Second:

$$\begin{aligned}\dot{\mathcal{V}}_1(\phi) &= -\mu \int_0^1 \phi_1^2(x) dx + (1 + \mu) \phi_1^2(1) - \phi_1^2(0) \\ &\quad + 2 \int_0^1 c(x) (1 + \mu x) \phi_1(x) \partial_x \phi_2(x) dx\end{aligned}\tag{5}$$

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Third:

$$\begin{aligned}\dot{\mathcal{V}}_{21}(\phi) &= -\mu \int_0^1 (\partial_x \phi_2)^2(x) dx + (1 + \mu) (\partial_x \phi_2)^2(1) - (\partial_x \phi_2)^2(0) \\ &\quad + 2 \int_0^1 (1 + \mu x) \frac{\lambda_2'(x)}{\lambda_2(x)} (\partial_x \phi_2)^2(x) dx\end{aligned}\quad (6)$$

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By introducing $\varepsilon > 0$, and selecting the positive scalars $\mu, \alpha_{20}, \alpha_{21}, \gamma$ as follows

$$\mu = \mathcal{O}(\bar{\nu}), \quad \gamma = \mathcal{O}(\varepsilon), \quad \alpha_{20} = \mathcal{O}\left(\frac{1}{\varepsilon}\right), \quad \alpha_{21} = \mathcal{O}\left(\frac{1}{\varepsilon^2}\right), \quad \text{for } \varepsilon \in \mathcal{O}(0)$$

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We end up with:

$$\dot{\mathcal{V}}(\phi) \leq -\delta \mathcal{V}(\phi), \quad \forall \delta < \min\left\{ \frac{\bar{\nu}\lambda_1}{1+\bar{\nu}}, \frac{\bar{\nu}\lambda_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\lambda_2}{(1+\bar{\nu})(1+\nu)} \right\}$$

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From (4), (5), (6) + **Young's inequality** and $\phi_2(1) = D_{22}\phi_2(0)$ we get:

$$\begin{aligned}\dot{\mathcal{V}}(\phi) \leq & (-\mu + (1 + \mu)\bar{c}\gamma) \|\phi_1\|_{L^2}^2 - \alpha_{20}\mu \|\phi_2\|_{L^2}^2 \\ & + \left(\frac{(1+\mu)\bar{c}}{\gamma} + \alpha_{21} \frac{\nu(2+\mu)-\mu}{1+\nu} \right) \|\partial_x \phi_2\|_{L^2}^2 \\ & + \left(\alpha_{20}(\mu - \bar{\nu}) + \left(\frac{1+\mu}{1-(1+\mu)D_{11}^2} \right) \frac{D_{12}^2}{D_{22}^2} \right) \phi_2^2(1) \\ & + \alpha_{21}(\mu - \bar{\nu})(\partial_x \phi_2)^2(1)\end{aligned}$$

By introducing $\varepsilon > 0$, and selecting the positive scalars $\mu, \alpha_{20}, \alpha_{21}, \gamma$ as follows

$$\mu = \mathcal{O}(\bar{\nu}), \quad \gamma = \mathcal{O}(\varepsilon), \quad \alpha_{20} = \mathcal{O}\left(\frac{1}{\varepsilon}\right), \quad \alpha_{21} = \mathcal{O}\left(\frac{1}{\varepsilon^2}\right), \quad \text{for } \varepsilon \in \mathcal{O}(0)$$

We end up with:

$$\begin{aligned}\dot{\mathcal{V}}(\phi) & \leq -\delta \mathcal{V}(\phi), \quad \forall \delta < \min\left\{ \frac{\bar{\nu}\lambda_1}{1+\bar{\nu}}, \frac{\bar{\nu}\lambda_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\lambda_2}{(1+\bar{\nu})(1+\nu)} \right\} \\ \Rightarrow \mathcal{V}(\phi(t; \phi_0)) & \leq e^{-\delta t} \mathcal{V}(\phi^0), \quad \forall \phi^0 \in X\end{aligned}$$

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Using the equivalence of state-space norm and Lyapunov induced norm on X ,

$$\|\phi(t; \phi^0)\|_X^2 \leq \kappa e^{-\delta t} \|\phi^0\|_X^2$$

SECOND CASE: $B \neq 0$

Assumption 2

The functions λ_2 , B and the matrix D satisfy:

$$\max(D_{11}^2, D_{22}^2) < 1 - \max \left\{ 2\overline{B_{11}}, \frac{\overline{\lambda_2'}}{\underline{\lambda_2}} + 2\overline{B_{22}} \right\}$$

Theorem 2

Let Assumption (2) holds and let

$$\nu_1 := 2\overline{B_{11}}(1 - 2\overline{B_{11}})^{-1}, \quad \nu_2 := \frac{\overline{\lambda_2'} + 2\underline{\lambda_2}\overline{B_{22}}}{\underline{\lambda_2} - \overline{\lambda_2'} - 2\underline{\lambda_2}\overline{B_{22}}}, \quad \bar{\nu} := \frac{1 - D_{22}^2}{D_{22}^2}.$$

Then, for any $\delta < \min\{\delta_1, \delta_2\}$ where $\delta_1 = \frac{\underline{\lambda_1}(\bar{\nu} - \nu_1)}{(1 + \nu_1)(1 + \bar{\nu})}$ and $\delta_2 = \frac{\underline{\lambda_2}(\bar{\nu} - \nu_2)}{(1 + \nu_2)(1 + \bar{\nu})}$, the origin of system (2) is δ -exponentially stable.

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$$\mathcal{W}_1(\phi) = 2 \int_0^1 (1 + \mu x) (B_{11}(x) \phi_1^2(x) + B_{12}(x) \phi_1(x) \phi_2(x)) dx.$$

$$\mathcal{W}_{21} = 2 \int_0^1 (1 + \mu x) B_{22}(x) \partial_x \phi_2^2(x) dx + 2 \int_0^1 (1 + \mu x) \frac{(\lambda_2 B_{22})'(x)}{\lambda_2(x)} \partial_x \phi_2(x) \phi_2(x) dx.$$

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Similarly to the first case, making use of Young's inequality and selecting the appropriate parameters, we achieve:

$$\dot{\mathcal{V}}(\phi) \leq -\delta \mathcal{V}(\phi), \quad \forall \delta < \min\left(\frac{\lambda_1(\bar{\nu} - \nu_1)}{(1 + \nu_1)(1 + \bar{\nu})}, \frac{\lambda_2(\bar{\nu} - \nu_2)}{(1 + \nu_2)(1 + \bar{\nu})}\right)$$

and the equivalence of norms ends the proof. ■

NUMERICAL RESULTS

ACADEMIC EXAMPLE

Consider system (2) with:

$$\lambda_1(x) = \begin{cases} 10, & x < \frac{1}{2} \\ -4(x - \frac{1}{2})^2 + 10 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\lambda_2(x) = \begin{cases} 10, & x < \frac{1}{2} \\ 4(x - \frac{1}{2})^2 + 10 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

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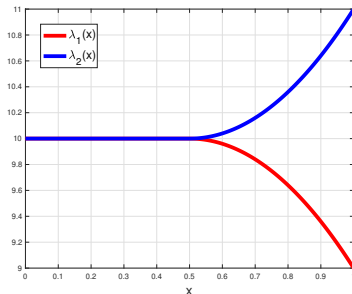
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ACADEMIC EXAMPLE

Regarding Theorem 1, the assumption $\max(D_{11}^2, D_{22}^2) = \frac{1}{4} < 1 - \frac{\lambda_2'}{\lambda_2} = \frac{3}{5}$ is satisfied.

Regarding Theorem 2, the assumption

$\max(D_{11}^2, D_{22}^2) = \frac{1}{4} < 1 - \max\left\{2\overline{B_{11}}, \frac{\overline{\lambda_2'}}{\lambda_2} + 2\overline{B_{22}}\right\} = \frac{4}{15}$ is satisfied.

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The system is discretized using an *Upwind* type explicit finite difference scheme:

$$\begin{cases} \frac{\phi_{1,i}^{n+1} - \phi_{1,i}^n}{\Delta t} = \lambda_{1,i} \frac{\phi_{1,i+1}^n - \phi_{1,i}^n}{\Delta x} + c_i \lambda_{2,i} \frac{\phi_{2,i+1}^n - \phi_{2,i}^n}{\Delta x} \\ \frac{\phi_{2,i}^{n+1} - \phi_{2,i}^n}{\Delta t} = \lambda_{2,i} \frac{\phi_{2,i+1}^n - \phi_{2,i}^n}{\Delta x} \end{cases}$$

Characteristic lines
through (x,t) plane

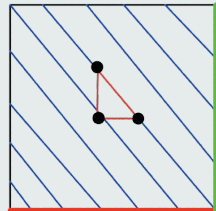
Setting:

$$\gamma_{1,i} = \frac{\lambda_{1,i} \Delta t}{\Delta x}, \quad \gamma_{2,i} = \frac{c_i \lambda_{2,i} \Delta t}{\Delta x}$$

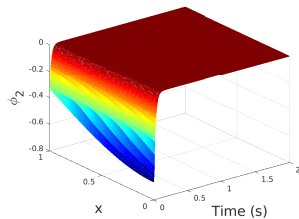
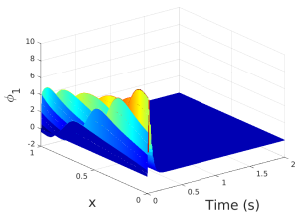
finally we get the numerical scheme:

$$\phi_{1,i}^{n+1} = (1 - \gamma_{1,i}) \phi_{1,i}^n + \gamma_{1,i} \phi_{1,i+1}^n - \gamma_{2,i} \phi_{2,i}^n + \gamma_{2,i} \phi_{2,i+1}^n$$

$$\phi_{2,i}^{n+1} = (1 - \gamma_{2,i}) \phi_{2,i}^n + \gamma_{2,i} \phi_{2,i+1}^n$$

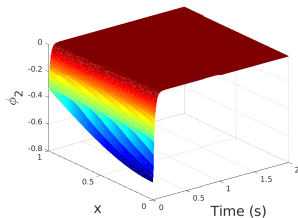
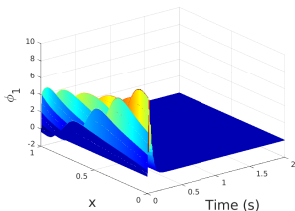


ACADEMIC EXAMPLE: SIMULATION

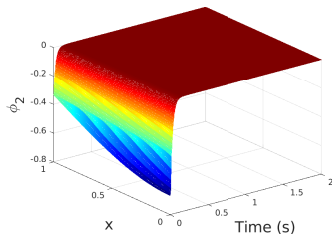
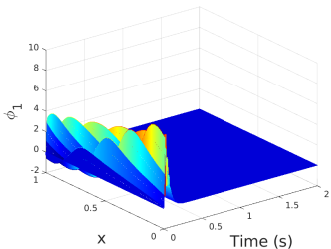


System (2) without source term: (left) state ϕ_1 / (right) state ϕ_2

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System (2) with source term: (left) state ϕ_1 / (right) state ϕ_2

ACADEMIC EXAMPLE: DECAY RATE

Recall δ -exponential stability:

$$\|\phi(t; \phi^0)\|_X^2 \leq \kappa e^{-\delta_{\text{th}} t} \|\phi^0\|_X^2, \quad \forall t \geq 0$$

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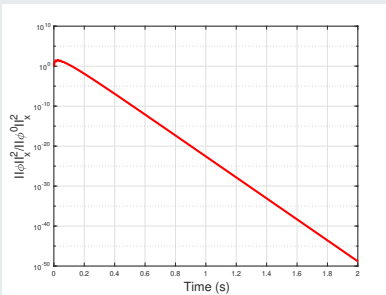
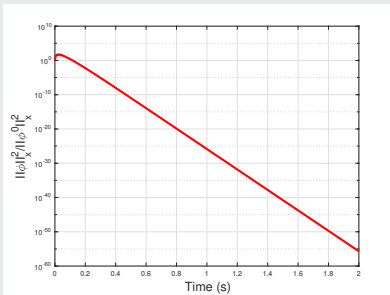
$$\Rightarrow \ln\left(\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}\right) \leq \ln(\kappa) - \delta_{\text{th}} t$$

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$\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}$ in log scale: (left) without source term / (right) with source term

ACADEMIC EXAMPLE: DECAY RATE

Without source term, theoretically, decay rate is given by
 $\min\left\{\frac{\bar{\nu}\lambda_1}{1+\bar{\nu}}, \frac{\bar{\nu}\lambda_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\lambda_2}{(1+\bar{\nu})(1+\nu)}\right\} = \frac{7}{2}$, whereas, numerically, decay rate is

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The difference is due to the conservatism introduced by the proposed Lyapunov analysis

CONCLUSIONS AND PERSPECTIVES

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 M.O. AMIRAT, V. ANDRIEU, J. AURIOL, M. BAJODEK and C. VALENTIN,
Stability analysis of a class of 2×2 triangular hyperbolic systems

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Tackle the case where the system **is not in cascade** form $\Rightarrow$ Choose a wider class of controllers (of Dynamical type ?)

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Conclude about the stability of the nonlinear system

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Tackle the case where the system **is not in cascade** form $\Rightarrow$ Choose a wider class of controllers (of Dynamical type ?)
Conclude about the stability of the nonlinear system
Apply to the Clarifier/Aeration tank system.

Thank you for your attention.
Questions ?